Lord of the Wings [2026]

Everybody Codes is possible thanks to:

Jane Street

Jane Street is a quantitative trading firm with offices worldwide.
We hire smart, humble people who love to solve problems, build systems, and test theories. Will our next great idea come from you?

Find out more

Quest 12: The Castastrophe

‹ Quest 11 Quest 13 ›

Part I

Story section

> We were already making our way towards the final portal when something behind me suddenly flashed red. I turned just in time to see the egg Frodoff was carrying glowing with a deep crimson light. The same red, mixed with gold, burned in Frodoff's eyes. Before I could say a word, Frodoff pushed me with strength no one should have possessed. I flew straight into one of the portals we had just passed.
>
> For a fraction of a second, I saw the portal come alive with thick black and crimson clouds. I knew that sight. It could mean only one thing. I had been thrown into the Abyss of Unmarked Recursion, the place where the most dangerous demons were banished to repeat their punishment for all eternity.
>
> And something was waiting for me there. Stackrog. I had sent him into this abyss myself during our previous expedition to destroy another cursed egg in Nulldor. That had been ages ago. Back then I was young, quick and strong. Now, standing before him once again, I was not nearly as certain that I could win. Fortunately, my memory had not failed me. I still remembered exactly how Stackrog could be defeated.

Stackrog attacked by hurling enormous stacks of magma towards me. Each stack broke apart in the air into individual magma blocks, forming a perfect rectangle before rushing towards me. That was the one thing working in my favour. Stackrog's attacks were powerful, but they were also remarkably orderly. **Every separate stack always formed its own rectangle,** which made reflecting the magma much easier than it might have been otherwise. I still remembered his first attack (your notes). Each `#` represents a magma block, while each `.` represents empty space.

Fortunately, the reflection spell I had used against him all those years ago came back to me. When cast, it travelled through a **straight horizontal or vertical line of neighbouring magma blocks**, reflecting every block it touched back towards Stackrog. The spell could travel only through magma. **It stopped as soon as it reached an empty space.** Each spell could therefore reflect one continuous horizontal or vertical line of magma blocks.

For example, a 3 by 6 stack:

```
######
######
######
```

could be reflected using `3` horizontal spells, one for each row:

```
------
------
------
```

or `6` vertical spells, one for each column:

```
||||||
||||||
||||||
```

I could keep casting, but that was exactly what Stackrog was counting on. Every spell consumed some of my energy, and I had no idea how long this fight might last. I had to make every spell count and **reflect all the magma blocks using as few spells as possible.**

Example based on the following notes:

```
........#.
.##.......
.##..####.
.##..####.
.....####.
.###......
```

This attack contains four separate stacks of magma.

The single magma block at the top requires `1` spell, which can be horizontal or vertical.

The 2 by 3 stack can be reflected using `2` vertical spells.

The 4 by 3 stack can be reflected using `3` horizontal spells.

Finally, the stack consisting of a single horizontal line requires only `1` spell.

The smallest total number of spells is therefore:

1 + 2 + 3 + 1 = `7`

The spells are shown below. For better readability, dots marking empty cells have been replaced with blank spaces. Dashes mark horizontal spells, and vertical bars mark vertical spells:

```
        |
 ||
 ||  ----
 ||  ----
     ----
 ---
```

What is the smallest number of spells needed to reflect all the magma blocks?

Your notes for this part:
Copy Open Download

Part I solved with answer: 217
Check your progress

Part II

Stackrog realised that his perfectly rectangular magma stacks were making my job far too easy. With an angry roar, he changed his strategy (your notes). Some of the incoming stacks still formed rectangles, but others now twisted in the air into **L-shaped formations.** Their arms could have different lengths and thicknesses, and the shapes could face in any direction.

Fortunately, my reflection spell still worked exactly as before. Each cast could reflect one continuous horizontal or vertical line of magma blocks, and it still could not cross an empty space.

Stackrog clearly hoped that the less convenient shapes would drain my energy faster. I had to adapt and once again **reflect every magma block using as few spells as possible.**

Example based on the following notes:

```
........#.
.##.......
.##..####.
.#...#....
.....#....
.####.....
```

This attack contains four separate magma stacks.

The single magma block at the top requires `1` spell.

The L-shaped stack on the left can be reflected using `2` vertical spells: one through its three-block column and one through its two-block column.

The larger L-shaped stack on the right can also be reflected using `2` spells: one horizontal spell along its long arm and one vertical spell along the other arm.

Finally, the horizontal stack at the bottom requires `1` spell.

The smallest total number of spells is therefore:

1 + 2 + 2 + 1 = `6`

```
...............................
.####....#####..#######........
.####....#####..#######........
.####....#####..#######........
.........#####.......##........
.........########....##........
.........########....##........
...............................
.######....#.....#.....###.....
...........#...........###.....
...........#......########.....
...........#......########.....
...........#......########.....
..................########.....
...............................
```

This attack contains several different kinds of magma stacks: rectangles, L-shaped formations with different arm thicknesses and orientations, straight lines, and a single magma block.

The 4 by 3 rectangle in the upper-left corner requires `3` spells.

The thick L-shaped stack beside it has a bounding box of 8 by 6. Covering its five-column vertical arm first would require 7 spells in total, but reflecting it row by row requires only `6`.

The L-shaped stack in the upper-right corner requires `5` spells.

The horizontal line, the vertical line, and the single magma block each require `1` spell.

Finally, the large L-shaped stack in the lower-right corner can be reflected with `6` horizontal spells. Covering its three-column vertical arm first and then the remaining part of its lower arm would instead require 7.

The smallest total number of spells is therefore:

3 + 6 + 5 + 1 + 1 + 1 + 6 = `23`

What is the smallest number of spells needed to reflect all the magma blocks?

Your notes for this part:
Copy Open Download

Part II solved with answer: 2043
Check your progress

Part III

Stackrog was getting frustrated. Rectangles had not worked, and neither had his twisted L-shaped formations. So he changed his attacks once again (your notes). Some magma stacks now formed **rectangles with a rectangular section missing from the stack**. The gaps could appear in different places and have different sizes, while the outer shape remained rectangular.

My reflection spell had not changed. Each cast still travelled through one continuous horizontal or vertical line of neighbouring magma blocks, and it still stopped as soon as it reached an empty space. This made the new stacks more troublesome. A gap could split a row or a column into separate parts, so choosing the right direction for each spell mattered even more.

The fight had already lasted far too long, and I could feel my energy fading. Once again, I had to **reflect every magma block using as few spells as possible.**

Example based on the following notes:

```
........#.
.##.......
.##..####.
.#...#..#.
.....####.
.###......
```

This attack contains four separate magma stacks.

The single magma block at the top requires `1` spell.

The L-shaped stack on the left requires `2` spells.

The stack on the right forms a 4 by 3 rectangle with a rectangular gap in its centre. It can be reflected using `4` spells.

Finally, the horizontal stack at the bottom requires `1` spell.

The smallest total number of spells is therefore:

1 + 2 + 4 + 1 = `8`

```
..............................
.##########....##.....#######.
.##########.##.##..#..#######.
.####..####.##.##.....#.#####.
.##########.##.#####..#.#####.
.##########.##........#######.
..............................
.##########..################.
.##########..#..............#.
.#######..#..#..............#.
.#######..#..################.
.##########..################.
..............................
```

This attack contains seven separate magma stacks, including rectangles, an L-shaped formation and several rectangles with gaps inside them.

The large stack in the upper-left corner forms a 10 by 5 rectangle with a 2 by 1 gap inside it. It requires `6` spells.

The L-shaped stack near the top requires `3` spells, the nearby 2 by 4 rectangle requires `2`, and the single magma block requires `1`.

The 7 by 5 stack in the upper-right corner contains a 1 by 2 gap and requires `6` spells.

The lower-left 10 by 5 stack contains a 2 by 2 gap and also requires `6` spells.

Finally, the wide stack in the lower-right corner is a 16 by 5 rectangle with a large rectangular gap inside it. Despite its size, it can be reflected using only `5` spells.

The smallest total number of spells is therefore:

6 + 3 + 2 + 1 + 6 + 6 + 5 = `29`

```
..............................
.############################.
.##...................###.###.
.##..################.###.###.
.##..######.........#.###.###.
.##..################.###.###.
.##...................###.###.
.##..####################.###.
.##..####################.###.
.##.......................###.
.############################.
..............................
```

A rectangular stack with a gap can contain other, separate magma stacks inside that gap. These stacks still have to be considered independently when choosing the spells.

For this attack, the smallest possible number of spells is `16`.

What is the smallest number of spells needed to reflect all the magma blocks?

Your notes for this part:
Copy Open Download

Part III solved with answer: 2824
Check your progress

Puzzle solved! Don’t stop now!

Post your solution, compare ideas, and help others grow on Reddit or join us for a chat on Discord

Bonus for Supporters!

This feature is available only to Supporters as a small "thank you" for supporting this event. Click the link below to **get all 100 versions of the input notes prepared for this Quest**, along with the answers! Please don't host it anywhere; especially please don't add it to your solutions repo. Thank you for being awesome! ❤

Download

© 2024-2026 Everybody Codes. All right reserved.
Terms of Use, Privacy Policy, Cookies and Tracking Policy
By using this website, you agree to these terms.